

**Anglican High School**  
**2020 Secondary 4 Preliminary Examination Mathematics Paper 2**

Answer **All** the questions

- 1**      **(a)**      **(i)**      Factorise  $2px - 2p + 3qx - 3q$  completely.

*Answer* ..... [2]

- (ii)**      Given that  $p$  and  $q$  are positive constants, find the value of  $x$  for which  
 $2px - 2p + 3qx - 3q = 0$ .

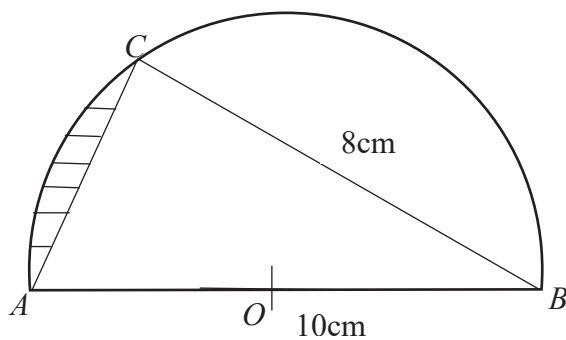
*Answer*  $x =$ ..... [1]

- (b)**      Simplify  $(-3p^2q^{-1})^2(p^{-2}q^2)^3$ , expressing your final answer in positive indices.

*Answer* ..... [2]

- 2 In the diagram,  $BC = 8$  cm and  $AB = 10$  cm is the diameter of the semi-circle.

Find



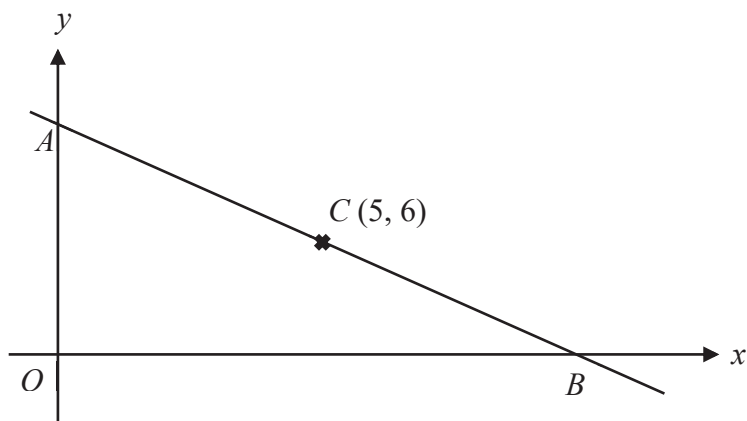
- (a) angle  $COA$  in radians,

Answer .....radians [2]

- (b) area of the shaded region.

Answer .....cm<sup>2</sup> [3]

- 3 In the diagram, not drawn to scale, point  $A$  lies on the  $y$ -axis and point  $B$  lies on the  $x$ -axis. The coordinates of  $C$  is  $(5, 6)$ .



- (a) Given that  $C$  lies on the line  $AB$  and that  $5OA = 3OB$ , show that the  $y$ -intercept of the line  $AB$  is 9.

*Answer*

- (b) Given that point  $D$  lies on the  $y$ -axis, state the coordinates of  $D$  such that triangle  $ACD$  is an isosceles triangle. [3]

*Answer* (....., ..... ) [1]

- (c) Given further that  $OCEA$  is a parallelogram, state the coordinates of the point  $E$ .

*Answer* (....., ..... ) [1]

- (d) Find the area of the parallelogram,  $OCEA$ .

*Answer* .....units<sup>2</sup> [2]